Pipeline Robustness & Graceful Degradation Report

Project: AI-Powered Crop Pest Prediction | Date: 2026-10-05 | Quality Assurance Lead

1. Stress Testing & Anomaly Resilience Protocol

Production machine learning systems deployed in rural agricultural domains encounter noisy sensor telemetry, missing inputs, unseen crop cultivars, and extreme anomalous weather events. The unified Scikit-Learn Pipeline was subjected to adversarial stress testing.

2. Adversarial Edge Case Verification Suite

Adversarial Test Case	Simulated Fault Condition	Architectural Guardrail	Status
Missing Numerical Values	Nulls/NaN in Temperature, Rainfall, Nitrogen	SimpleImputer (median fitted on training split)	PASSED
Missing Categorical Values	Nulls/NaN in District or Crop_Name	SimpleImputer (most_frequent fitted on train)	PASSED
Unknown District	District = 'UnseenDistrict_X99'	OrdinalEncoder(handle_unknown='use_encoded_value', unknown_value=-1)	PASSED
Unknown Crop Cultivar	Crop = 'ExoticDragonFruit_Alpha'	OrdinalEncoder handles unseen category as -1	PASSED
Unknown Soil Classification	Soil_Type = 'Volcanic_Pumice_Soil'	OrdinalEncoder handles unseen category as -1	PASSED
Extreme Microclimatic Extremes	Temp=62.5°C, Rain=1200mm, pH=12.8, Wind=120km/h	StandardScaler bounds + Decision tree threshold monotonicity	PASSED

3. Production Safety Conclusion

100% of adversarial and boundary queries were processed without server exceptions or unexpected terminations. Predictions gracefully degraded to regional prior probabilities when location and crop data were unmapped.